

Non-existent entries in certain tags that are not required causes validation error

MC-275790 Data Packs

Status	Resolved / Fixed
Affected	24w34a, 24w35a, 24w36a, 24w37a, 24w38a, 24w39a
Fixed in	24w40a
Reported	2024-08-21

STEPS TO REPRODUCE

1. Launch Minecraft 24w34a.
2. Create a data pack that does not define the biome ID `test:test`.
3. Add a biome tag at `/tags/worldgen/biome/<tag\_name>.json` with the following contents:
```json  
{  
 "values": [  
 "minecraft:plains",  
 {  
 "id": "test:test",  
 "required": false  
 }  
 ]  
}  
```
4. Open Singleplayer and create a new world with default settings.
5. Enable the data pack in the world's data pack selection screen.
6. Confirm the data pack selection and attempt to load the world.
7. Observe that the data pack fails validation and the game attempts to launch the world in Safe Mode.
8. Check the output log and observe registry loading errors identifying `test:test` as an unbound or missing biome entry.

OBSERVED BEHAVIOUR

Actual behavior

When a data pack contains a biome, structure, enchantment, or loot table tag with an object entry whose `required` property is set to `false`, the game still treats a missing referenced ID as an unbound registry entry.

For example, loading the following biome tag in 24w34a:

```
```json
{
 "values": [
 "minecraft:plains",
 {
 "id": "test:test",
 "required": false
 }
]
}
```
```

fails during registry loading if `test:test` is not defined by any loaded data pack. The log reports an error such as:

```
```text
java.lang.IllegalStateException: Unbound values in registry
ResourceKey[minecraft:root / minecraft:worldgen/biome]: [test:test]
```
```

The data pack is consequently reported as invalid, and the world creation screen offers or attempts to use Safe Mode. The missing ID is listed in the registry loading errors even though its tag entry is explicitly marked as not required.

This behavior differs from 1.21.1, where the same tag validates successfully and the missing, non-required entry is ignored.

EXPECTED BEHAVIOUR

When a biome, structure, enchantment, or loot table tag contains an entry marked `"required": false` whose ID is absent from all loaded data packs, the missing entry should be ignored. The tag should still validate successfully, retaining any entries that exist, and the game should allow the data pack to load normally without reporting registry errors or prompting to launch in Safe Mode.

ENVIRONMENT

Minecraft Java Edition Versions: 24w34a, 24w35a, 24w36a, 24w37a, 24w38a, 24w39a (includes snapshots)

ORIGINAL DESCRIPTION

If a biome, structure, enchantment, or loot table tag is included in a data pack, which includes an entry with required set to false, and that biome/structure/enchantment/loot_table ID is not present in any data packs, the data pack fails to validate and the game tries to launch in Safe Mode. The output log produces this error (example with biome tag):

deobf_latest.log

[Render thread/ERROR]: Registry loading errors:

> Errors in registry minecraft:root:

>> Errors in element minecraft:worldgen/biome:

```
java.lang.IllegalStateException: Unbound values in registry ResourceKey[minecraft:root /
minecraft:worldgen/biome]: [test:test]
```

```
at net.minecraft.core.MappedRegistry.net.minecraft.core.Registry freeze()(MappedRegistry.java:333)
```

```
at net.minecraft.resources.RegistryDataLoader.<void>
```

```
lambda$load$6(java.util.Map,net.minecraft.resources.RegistryDataLoader$Loader)(RegistryDataLoader.java:188)
```

```
at java.base/java.lang.Iterable.forEach(Iterable.java:75)
```

```
at net.minecraft.resources.RegistryDataLoader.net.minecraft.core.RegistryAccess$Frozen load(net.minecraft.resour
ces.RegistryDataLoader$LoadingFunction,java.util.List,java.util.List)(RegistryDataLoader.java:185)
```

```
at net.minecraft.resources.RegistryDataLoader.net.minecraft.core.RegistryAccess$Frozen load(net.minecraft.server
.packs.resources.ResourceManager,java.util.List,java.util.List)(RegistryDataLoader.java:164)
```

```
at net.minecraft.resources.RegistryDataLoader.<void> lambda$load$2(net.minecraft.server.packs.resources.Resource
Manager,net.minecraft.resources.RegistryDataLoader$Loader,net.minecraft.resources.RegistryOps$RegistryInfoLo
okup)(RegistryDataLoader.java:164)
```

```
at net.minecraft.server.WorldLoader.java.util.concurrent.CompletableFuture load(net.minecraft.server.WorldLoader
$InitConfig,net.minecraft.server.WorldLoader$WorldDataSupplier,net.minecraft.server.WorldLoader$ResultFactory,j
ava.util.concurrent.Executor,java.util.concurrent.Executor)(WorldLoader.java:41)
```

```
at net.minecraft.client.gui.screens.worldselection.CreateWorldScreen.<void> applyNewPackConfig(net.minecraft.serv
er.packs.repository.PackRepository,net.minecraft.world.level.WorldDataConfiguration,java.util.function.Consumer)(
CreateWorldScreen.java:574)
```

```
at net.minecraft
```

MANUAL RESULT

| | |
|------------------|--------------------|
| Version used | |
| Reproduced? | YES / NO / PARTIAL |
| Crash file | |
| Notes | |
| | |
| Tested by / date | |

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-275790>